Supply Chain Analytics

Data-Driven Insights Across Three Phases

Phase 1

EDA & Data Cleaning

Phase 2

Predictive Modeling

Phase 3

Advanced ML & Explainability

T E A M C

EVOASTRA VENTURES
DIGITAL SOLUTIONS

Phase 1 | Dataset Overview & Cleaning

PHASE 1

180,519 rows

Original dataset size

53 columns

Features available

~99.7% missing

Product Description — dropped

~86% missing

Order Zipcode — dropped

Cleaning Steps

1. Dropped high-missing columns (>50%)
2. Removed rows with minimal missing values
3. Standardised column names (lowercase)
4. Converted date columns to datetime format
5. Created delivery_days feature
6. Applied IQR-based outlier detection
7. Created binary delay_flag target variable

Phase 1 | Key EDA Findings

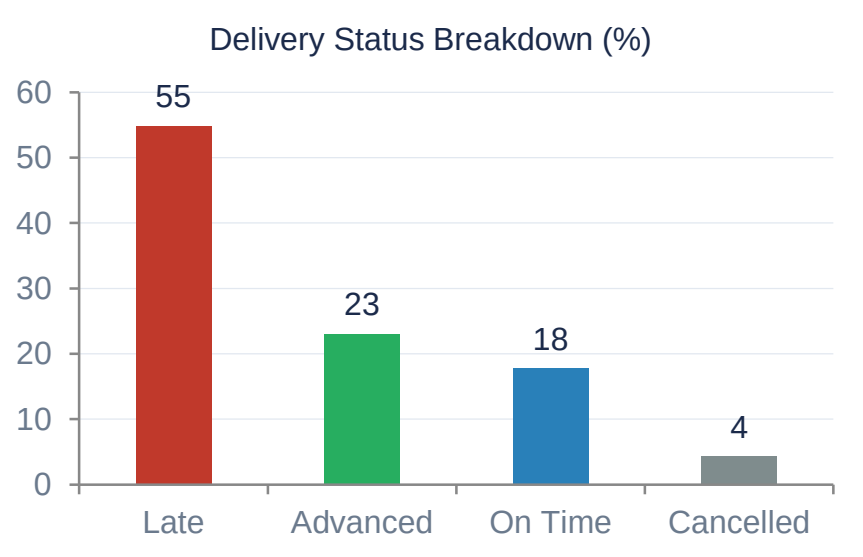
PHASE 1

54.8%
Late Deliveries

$36.78M
Total Sales

$21.97
Avg Profit/Order

3.47 days
Avg Delivery Time



Other Insights

- Top categories: Cleats, Men's Footwear, Women's Apparel
- Computers generate highest average profit per order
- Caguas accounts for the highest city-level revenue
- Delays are consistent across departments and regions — an operational issue
- Sales vs Profit: positive but weak (r = 0.13); high discounts erode margins
- Monthly sales stable 2015–2017 with a mid-2017 peak

Phase 1 | KPIs & Correlation Summary

PHASE 1

Sales vs Profit

Weak positive — sales alone don't guarantee profit

+0.13

Discount Rate vs Profit

Higher discounts marginally reduce profitability

−0.02

Delivery Days vs Real Shipping

Near-perfect overlap — one can be dropped in ML

+1.0

Delivery Days vs Scheduled

Moderate — scheduling directly affects delivery performance

+0.53

Phase 1 Outcome: Clean, feature-rich dataset ready for modeling. Key target variable — delay_flag — identified for Phase 2.

Phase 2 | Statistical Testing

PHASE 2

T-Test

Debit vs Cash — Delivery Days

No statistically significant difference in delivery time between debit and cash orders.

$p = 0.0716$ → Not significant

ANOVA Test

Delivery Days across Transaction Types

Delivery days vary significantly across debit, transfer, payment, and cash types. Transaction type influences supply chain efficiency.

$F = 7.97$, $p < 0.05$ → Significant

Chi-Square Test

Transaction Type vs Delivery Status

A meaningful association exists between transaction type and delivery outcome — certain types are more delay-prone.

$p < 0.05$ → Significant

Phase 2 | Feature Engineering & Model Building

PHASE 2

Feature Engineering

Date Features

Extracted day, month, weekday from order dates

Lag Features

lag_1 (previous sales) captures demand momentum

Rolling Mean

7-day rolling average smooths fluctuations

Label Encoding

Categorical variables (shipping_mode, market…) encoded numerically

Standard Scaling

All features scaled for regression model stability

Models Built

Linear Regression
Baseline

Ridge Regression
Regularised (L2)

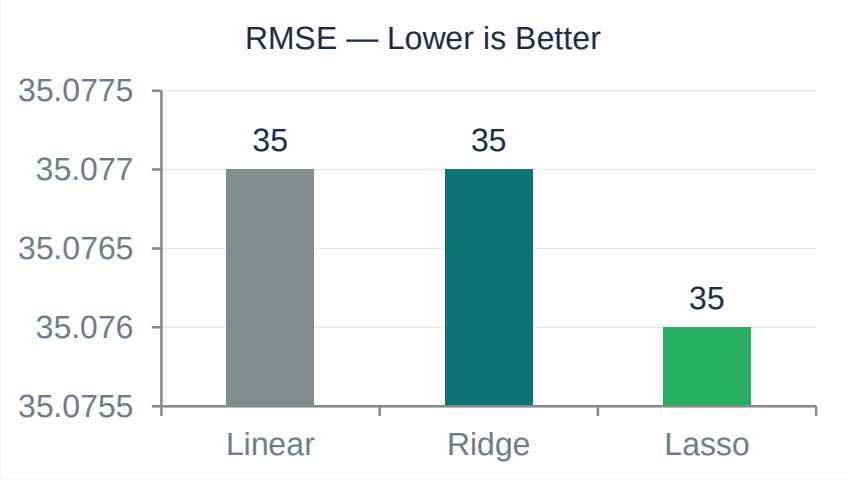
Lasso Regression
Best performer

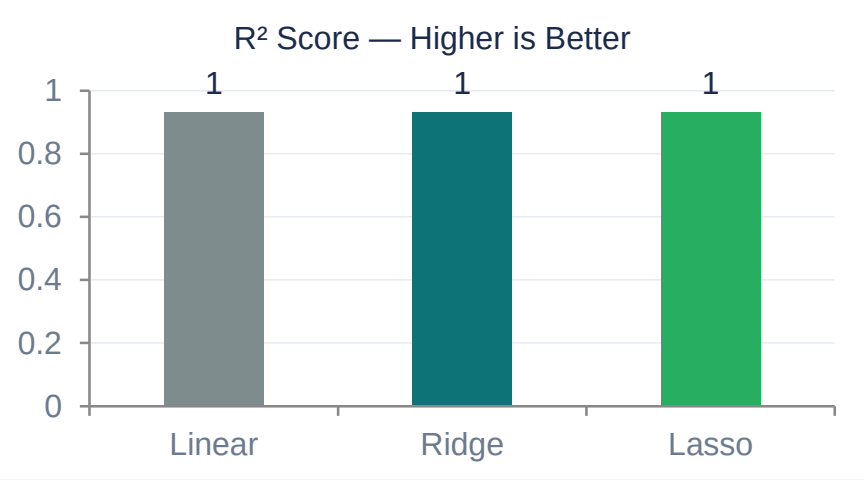
ARIMA (5,1,0)
Time series forecast

SARIMA (1,1,1)×(1,1,1,7)
Seasonal forecast

Phase 2 | Model Evaluation & Results

PHASE 2





Time Series Forecast Results

ARIMA: predicted range ~176–191 units | SARIMA: ~183–215 units (captures weekly seasonality)

Best Model: Lasso Regression — R^2 = 0.9306 | RMSE = 35.076 | MAE = 18.23

Phase 3 | Advanced ML Models & Hyperparameter Tuning

PHASE 3

Models Trained

Random Forest

n_estimators: 100 | max_depth: 10
RMSE: 22.14 | $R^2 = 0.964$
Best params via GridSearchCV (3-fold)

XGBoost

n_estimators: 100 | learning_rate: 0.1
RMSE: 20.87 | $R^2 = 0.967$
Best params via GridSearchCV (3-fold)

Tuning & Validation

GridSearchCV

Exhaustive grid search over n_estimators, max_depth, learning_rate, min_samples_split

3-Fold Cross-Validation

KFold applied on both RF and XGBoost for robust out-of-sample estimates

Feature Set

13 features: shipping/delivery days, quantity, date parts + 7 encoded categoricals

Baseline vs Tuned

Phase 2 Lasso R^2=0.93 → Phase 3 XGBoost R^2=0.967 — meaningful improvement

Phase 3 | SHAP Explainability & Project Summary

PHASE 3

SHAP Explainability (XGBoost)

- Summary Plot identifies global feature impact across all predictions
- Bar Plot ranks features by mean absolute SHAP value
- Force Plot explains a single prediction locally
- Top drivers: days_for_shipping, delivery_days, order_item_quantity
- Confirms model uses logistics-relevant signals, not noise

Phase-wise Progress

Phase	Model	R^2	RMSE
1	Baseline EDA	—	—
2	Lasso Regression	0.9306	35.08
3	Random Forest	0.964	22.14
3	**XGBoost (best)**	**0.967**	**20.87**

Phase 3 Outcome: XGBoost achieved R^2 = 0.967 — best model across all phases. SHAP confirmed delivery days and shipping speed as the primary sales drivers.

Key Takeaways

Phase 1 — Data Quality & EDA

- Clean dataset of 180K+ orders
- 55% late delivery rate — key operational gap
- IQR outlier handling, delay_flag target created

Phase 2 — Modeling & Forecasting

- ANOVA & Chi-square confirmed transaction type impacts delivery
- Lasso Regression achieved R^2 of 0.93
- SARIMA captured seasonal demand patterns

Phase 3 — Advanced ML & Explainability

- XGBoost achieved best R^2 of 0.967
- SHAP confirmed delivery days & shipping speed as top drivers
- Random Forest $R^2 = 0.964$ validated ensemble approach